



Question 1:

Marks: 8

A) Mean Shift is a popular method used in computer vision for problems such as image segmentation and object tracking. **[4 Marks]**

- i. Describe the main steps of how Mean Shift is used to perform image segmentation, including the parameter(s) that must be set in order to use Mean Shift.

Solution:

Lec 2 slides 41 -42

Parameters: Window size

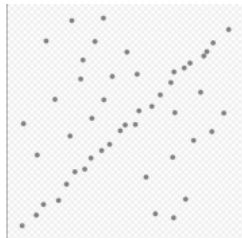
- ii. Comment on the sensitivity of the algorithm to its parameter value(s); i.e., what are the effects of setting the parameter(s) to values that are too big or too small.

Solution:

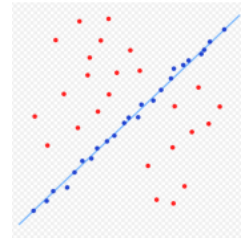
If window size too big: low time complexity until convergence but low accuracy

If too small: high time complexity for convergence but high accuracy.

B) Given the following 2-D point cloud, it is supposed to fit an optimal line model to it. [4 Marks]



(a) Original point cloud



(b) Line model after fitting

- i. Describe how you will use **RANSAC** algorithm for the aforementioned task.

Solution:

Lec 6 slides 17-22

- ii. Assume that there are 60% inliers; compute the minimal number of RANSAC iterations needed to get, with probability 95%, at least one random sample that is free from outliers.

Solution:

$$N > \log(1-p) / \log(1-(1-e)^s)$$

$$P = 0.95 \quad e = 0.4, \quad s = 2$$

$$N > \log(1-0.95) / \log(1-(1-0.4)^2)$$

$$N > \log(0.05) / \log(1-0.36)$$

$$N > \log(0.05) / \log(0.64)$$

$$N > 6.712$$

$$N = 7$$

Question 2:

Marks: 12

A) What feature detector would you use to propose a set of probable correspondences between two images of the same object, but taken with a different rotation and zoom setting?

SIFT

Explain only the step(s) of how this feature detector is set to be rotationally invariant? [4 Marks]

Solution:

Lec 5 slides 19 - 24

B) Explain a scheme for accomplishing skin color recognition in images with the following emphasis: [4 Marks]

- i. Explain what kind of low-level feature extraction you can use.

Solution:

The Hue/color value can be used as a feature x that represents the skin color.

- ii. At the level of classifier, explain how the system can use Bayesian Framework to incorporate expert knowledge such as the skin color frequencies in images.

Solution:

A Bayesian classifier can be used as some distance or similarity metric to determine if the input data is a skin or not.

It computes a probability $P(C_k|x)$ for the skin class C_k given the measured feature data x , in terms of the likelihood $P(x|C_k)$

$$P(C_k|x) = P(x|C_k)P(C_k) / P(x)$$

Expert knowledge such as relative skin frequency can be embedded into this classifier in the Bayesian framework as the term $P(C_k)$ in the above expression.

C) Consider the following 2-class binary problem [4 Marks]

$$C_y = -1 = \{(-3, -1), (-3, 1), (3, -1), (3, 1)\}$$

$$C_y = 1 = \{(-1, -1), (-1, 1), (1, -1), (1, 1)\}$$

Using the error and weight update rules of the AdaBoost algorithm:

- i. Construct the first weak learner and re-update the data points' weights.

Solution:

The decision boundary will be in the middle as all data points have the same weights $w_i = 1/n = 1/8 = 0.125$. Weak learners are highlighted on the figure.

Weight update equations:

$$w_i^m = \begin{cases} w_i^{m-1} * \frac{1 - \text{err}}{\text{err}} & \text{for weak learners} \\ w_i^{m-1} & \text{otherwise} \end{cases}$$

$$\text{err} = \frac{\sum_{i=1}^n w_i \mathbb{I}\{y_i \neq g_m(x_i)\}}{\sum_i w_i}$$

$$\text{err} = (0.125 * 4) / 1 = 0.5$$

$$w_i = 0.125 * (1 - 0.5 / 0.5) = 0.125 \quad \text{for all weak learners.}$$

- ii. Do you think this solution is sufficient to obtain a perfect classification? If not, explain why and give your opinion on what is needed for the problem to be correctly classified.

Solution:

This solution is not sufficient as all weights will remain unchanged.

The classifier function will be always 0 as long as the weak learners weights are equal on both sides of the decision boundary. We need at least 3 weak learners to correctly classify the problem as the third one plays the role of a bias.

With Best Regards,
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